SMARTDESK

Design and Evaluation of an Intelligent IT Support Ticket Classification System

A Reproducible Proof-of-Concept Study

Research area: IT Support, NLP, and Applied Machine Learning

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Synthetic-data proof of concept; not field validation

Abstract

Comparison of keyword rules, TF-IDF + Naive Bayes, and TF-IDF + Logistic Regression on 1,000 synthetic IT tickets. Observed accuracies were 100.0%, 100.0%, and 100.0%. These scores apply to generated examples only.

1. Objectives

Create a five-class benchmark, compare three lightweight methods, report standard metrics, and document limitations.

2. Methodology

1,000 balanced synthetic examples; stratified 80/20 split, seed 42; five categories: Network, Hardware, Software, Access, Storage. TF-IDF uses unigrams and bigrams. Metrics: accuracy, macro precision, recall, F1.

3. Results

The metrics below were generated by running the included experiment on the synthetic benchmark; they are not real-world results.

Model	Accuracy	Macro P	Macro R	Macro F1	Seconds*
Keyword Rules	1.000	1.000	1.000	1.000	0.001
TF-IDF + Naive Bayes	1.000	1.000	1.000	1.000	0.015
TF-IDF + Logistic Regression	1.000	1.000	1.000	1.000	0.023

*Local timing; machine-dependent.

4. Discussion

TF-IDF models score higher on generated examples, likely because category-specific words recur in templates. Scores may reflect generator artifacts rather than real-world generalization.

5. Limitations

Synthetic data; shared template process; one split; limited taxonomy; no human comparison or real ticket evaluation; no confidence intervals; machine-dependent runtime.

6. Ethics and Future Work

Use authorized de-identified real tickets, annotation guidelines, a locked test set, per-class error analysis, and human review. Do not treat these synthetic results as evidence of operational accuracy.

7. Conclusion

Observed accuracies were 1.000, 1.000, and 1.000. This is a reproducible proof of concept, not a validated production system.

References

Manning et al. (2008), Introduction to Information Retrieval. McCallum & Nigam (1998), A comparison of event models for Naive Bayes text classification. Pedregosa et al. (2011), Scikit-learn: Machine Learning in Python, JMLR 12, 2825–2830.

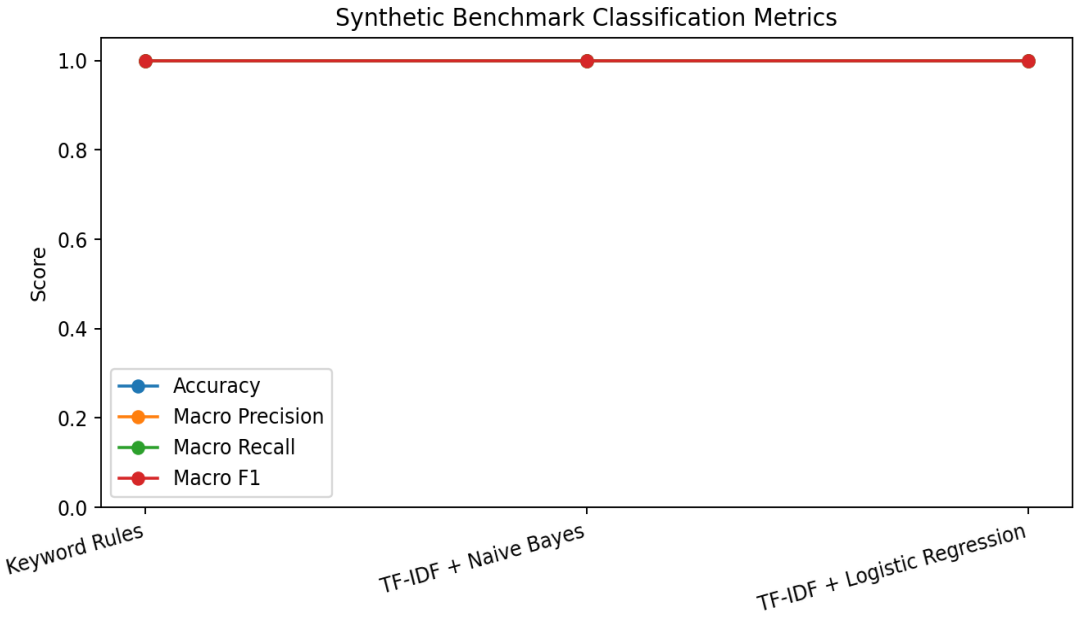


Figure 1. Classification metrics.

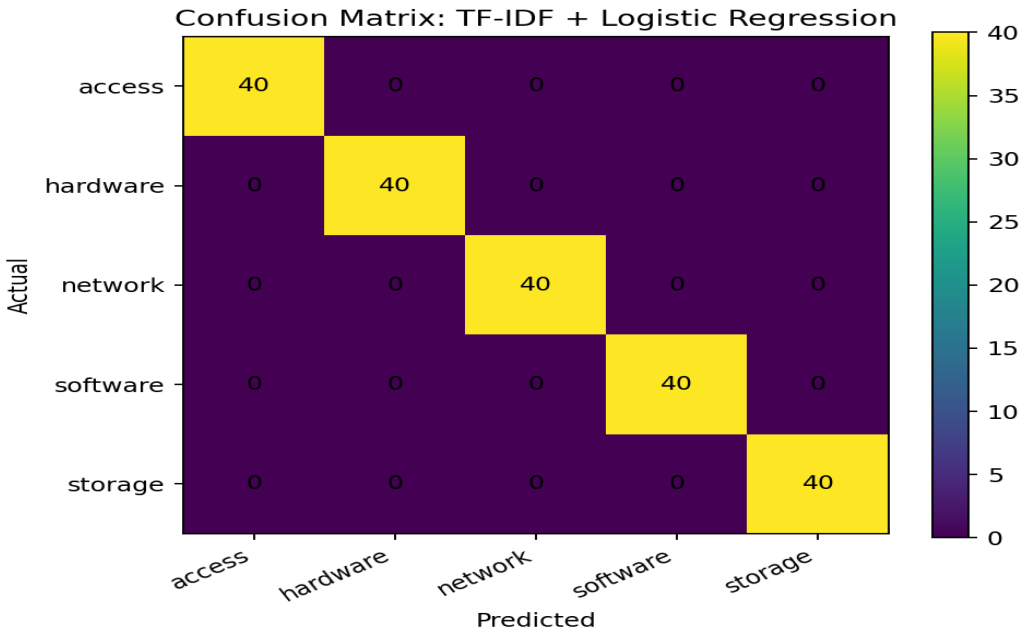


Figure 2. Confusion matrix.